Case E

Mr. Liam O'Connor, 19M — Cystic Fibrosis (Pulmonary Tune-up) | FEV■ decline · Increased purulent sputum

Target: 70%+ | Cardiorespiratory Physiotherapy

■ PROBLEM LIST

- Impaired mucociliary clearance → retained secretions
- Airflow obstruction (↓ FEV■)
- Increased mucus viscosity (CFTR dysfunction)
- Reduced exercise capacity / deconditioning
- Minor haemoptysis (fragile inflamed bronchial mucosa)
- Unable to complete home PEP without exhaustion
- Exercise tolerance reduced (dyspnoea-limited)
- Difficulty maintaining therapy around university schedule
- Risk of further FEV■ decline if adherence not addressed

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

- Core defect: Mutations in the CFTR gene produce dysfunctional or absent chloride channels on airway epithelium.
- This leads to dehydrated, hyperviscous airway surface liquid — mucus that cannot be cleared by normal mucociliary action.
- Infection-inflammation cycle: Retained mucus plugs obstruct small and medium airways, creating a nidus for chronic infection — predominantly Pseudomonas aeruginosa .
- Neutrophil-dominated inflammation releases elastase and proteases that destroy airway walls, driving bronchiectasis, air trapping, and progressive obstructive lung disease.
- FEV■ decline mechanism: Each exacerbation accelerates bronchiectatic remodelling, reducing elastic recoil and increasing resistance.

■ DIFFERENTIAL DIAGNOSES

1. Allergic Bronchopulmonary Aspergillosis (ABPA)

- Common CF complication (prevalence 2–15%)
- Presents with worsening airflow obstruction, increased mucus, and constitutional symptoms
- Streaky haemoptysis and elevated total IgE may occur

2. Non-CF Bronchiectasis (e.g., post-infective)

- Shares features: chronic productive cough, purulent sputum, recurrent chest infections
- Similar airflow obstruction pattern and exercise intolerance
- Haemoptysis can occur in bronchiectasis of any aetiology

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Spirometry — FEV■ / FVC (Pre & Post Bronchodilator)
What it measures Degree of airflow obstruction; tracks decline from personal best; assesses bronchodilator reversibility How to perform Patient seated, noseclip applied. Maximal inspiration then rapid, forceful, sustained expiration for ≥6 seconds. Repeat 3 acceptable manoeuvres;

Six-Minute Walk Test (6MWT) or Modified Shuttle Walk Test
What it measures Functional exercise capacity; oxygen desaturation on exertion; baseline for exercise rehabilitation How to perform 10- metre flat corridor (6MWT) or pre-recorded beep levels (MSWT). Monitor SpO■, HR, and Borg (0–10) at each minute. Record distance achieved. Stop

OPEP Technique Assessment (Aerobika / Acapella)
What it measures Accuracy of current home airway clearance technique; identifies technique errors contributing to inadequate secretion mobilisation How to perform Ask Liam to demonstrate his OPEP routine. Observe: upright posture, tidal breathing through device (not forced), resi

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–2 (Inpatient Tune-up)

- Sequencing matters: Always use mucolytics (DNase, hypertonic saline) 30–60 min before airway clearance — this dramatically improves sputum mobilisation and makes every session more effective.
- Adherence strategy: Two sessions daily is the evidence-based minimum; we will map sessions to your lecture timetable — a morning and evening 20-minute routine is realistic even around a full university day.
- Exercise is treatment, not extra: Aerobic exercise augments mucociliary clearance and slows FEV■ decline — it is a core part of CF management, not optional.
- Haemoptysis protocol: Minor streaking during coughing is common in CF. If blood volume increases significantly, you feel unwell, or it occurs outside coughing, stop all airway clearance and contact the team immediately.

■ Long-term — Weeks 4–12 (Outpatient / Self-Management)

- CF Pulmonary Rehabilitation: Supervised aerobic + resistance programme 2–3× per week; target ≥150 min moderate-intensity exercise per week

• Adherence tools: CF diary app (e.g., myCF) to log sessions; phone reminders mapped to university timetable; involve CF nurse specialist
• Airway clearance independence: Self-managing OPEP + ACBT; selects appropriate technique based on daily secretion volume
• Annual review: Spirometry tracked against personal best; written exacerbation action plan in place
• Discharge criteria: FEV■ ≥85% personal best · SpO■ ≥93% at rest · Independent airway clearance ×2/day · Functional exercise capacity at pre-admission level · University schedule plan documented